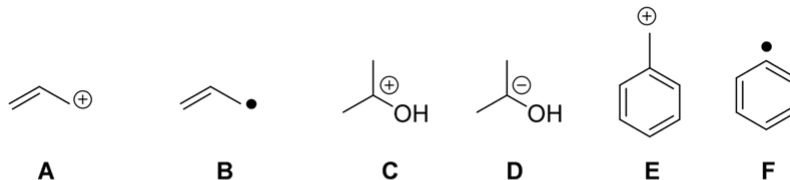
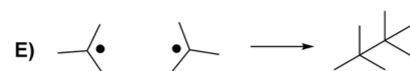
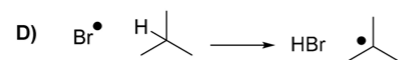
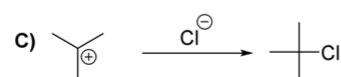
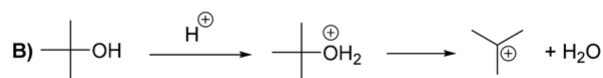
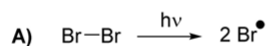


QX4004 – Engineering Chemistry
Resonance and Free Radicals Problem Set

- 1) For the following species, show the structural formula (show all the carbons, hydrogens and heteroatoms), determine the hybridisation for all the carbons, and write resonance forms when suitable.

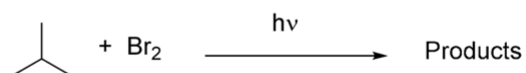


- 2) Use curly arrows, and lone pairs where appropriate, to complete the following diagrams:



- 3) Why does the stability of free-radicals follow the trend: tertiary > (more stable than) secondary > primary > methyl?

- 4) Consider the following reaction:



- Draw the structure of all possible products, identify the main product and explain why it is the main product.
- Write a mechanism for the reaction leading to the main product (A "reaction mechanism", more often simply "mechanism" shows, through the use of curly arrows, how bonds are broken and formed, in other words, is a way of showing the reaction pathway).